From Static to Contextual

Distributional Semantics and What's Inside BERT

Two independent implementations, one set of conclusions

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Common presentation

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The problem

Common data: UD English EWT – 12,544 train / 2,001 dev / 2,077 test sentences; 204,578 training tokens. Gold dependency heads support the syntax experiments.

Task 1 – baseline

PPMI + SVD vectors from EWT training text: one fixed vector per word type. **Distributional context**

Task 2 – reproduction

Hewitt–Manning structural probe on frozen `bert-base-uncased`; inspect layers 1–12 to find the strongest syntax layer ℓ^*. **Contextual structure**

Task 3 – combine

Average layer-ℓ^* representations over word occurrences to form static BERT vectors; compare with Task 1. **Pooled representation**

Two complementary evaluations

Meaning: WordSim-353 Spearman correlation and Google analogy accuracy.

Syntax: Build an MST from pairwise word-vector distances; compare its edges with the gold dependency tree (UUAS).

Research question

Can pooling from the syntax-encoding layer improve static word meaning without harming syntax – and which findings remain stable across two independent implementations?

Two implementations: what is shared, what differs

Shared experimental core

- Identical EWT train/dev/test splits.
- Same WordSim-353 and Google analogy benchmarks.
- Frozen `bert-base-uncased`; word vector = mean of its WordPiece vectors.
- Distance and depth structural probes; L1 losses.
- Select best layer using development data.

	A (Y. D. Babu)	**B** (T. Saha)
T1 vocabulary	All 16,654 types	8,865 types (count ≥ 2)
WS "all 353"	OOV pair $\rightarrow$ median similarity In-vocabulary;	OOV pair $\rightarrow 0$ All words;
20-sentence MST	punctuation excluded	punctuation included
Larger MST set	660 eligible sentences	1,926 test sentences
Probe learning rate	$\times 0.1$ **decay on dev plateau**	**Constant** 10^{-3}
T3 vocabulary	All word types	T1 vocabulary (+ all-type variant)
Anisotropy correction	All-but-the-top ($D = 7$)	Mean-centering; PCA-100
Uncertainty reporting	Bootstrap CIs and paired tests	Micro/macro metrics; full test set

The differences that matter most

Vocabulary

16,654 vs. 8,865 types changes coverage and candidate sets.

Optimization

The learning-rate schedule changes probe convergence and the best rank.

Evaluation protocol

OOV policy, punctuation, and sampled sentences shift reported scores.

Code organisation of the two implementations

A: author-folder/code/

`common/`	Data, MST, statistics, shared evaluation
`task1/build_ppmi_svd.py`	Co-occurrence → PPMI → SVD
`task2/extract_bert.py`	13-layer feature cache
`task2/train_probes.py`	78 probes; learning-rate decay
`task3/pool_static.py`	Mean vector per word type
`task3/evaluate_task3.py`	Benchmarks and paired tests
`run_all.py + Colab`	One-command run

B: Codes/ + results/

`common.py`	EWT .conllu, benchmarks, MST
`Task1.py`	Co-occurrence, PPMI, sparse SVD
`Task2.py`	Features, probe loss, chunked probe training
`Task3.py`	Static vectors, centering/PCA, sweep
`results/`	JSON/CSV outputs and figures

Same repository, clearly separated

`github.com/trishitasaha/NLP_Assignment2`: **B**'s implementation is at the root (`Codes/`, `results/`); **A**'s implementation is in its own folder; the common report and slides are in `Combined/`.



Both codebases separate the three tasks, while sharing the same data and benchmark definitions.

Cross-check: same settings ⇒ same numbers

Quantity	**A** re-scored in **B**'s setting	**B** reported
Types with count ≥ 2	8,865	8,865
Analogy questions, all words count ≥ 2	3,406	3,406
T3 WS-353 ρ on the same 203 pairs	**0.648**	**0.648**
T3 WS-353 ρ, all 353, missing $= 0$	**0.288**	**0.288**
T3 WS-353 ρ, 248 covered pairs	0.577	0.577
T3 analogy, covered / all questions	67.7% / 19.8%	67.7% / 19.8%
T1 WS-353 ρ on the 203 pairs	0.171	0.189

What the cross-check establishes

- The Task 3 pipelines are equivalent: matched figures agree to three decimals.
- Restricting to 8,865 types changes the analogy candidate set and pair coverage.
- Task 1 differs because **B** removes singleton words before SVD.

Why this control is important: a fair implementation comparison must hold vocabulary, scored item set, missing-value policy, and analogy candidates fixed. Otherwise a score change can come from evaluation coverage rather than model quality.

Matched evaluation set: 8,865 word types | 203 common WordSim pairs | 3,406 common analogy questions.

Task 1: both find weak, data-limited vectors

Task 1: PPMI–SVD	A	B
WS-353 ρ, covered pairs	0.104 (248)	0.189 (203)
WS-353 ρ, all 353	0.048 (median)	0.024 (zero)
Analogy accuracy, covered	0.51%	0.85%
MST UUAS, 20 sentences	20.5%	16.7%
random / linear-chain baseline	15.2 / 43.2	13.1 / 35.5
MST UUAS, larger set	24.8% (660)	17.8% (1,926)

What the baseline is learning

Co-occurrence within a symmetric ± 5 window captures topical partners. The neighbour examples confirm this: *monday* $\rightarrow$ Friday / July / Tuesday; *good* $\rightarrow$ nice / food / great. The small corpus still gives sparse counts and weak similarity estimates.

Agreement across runs

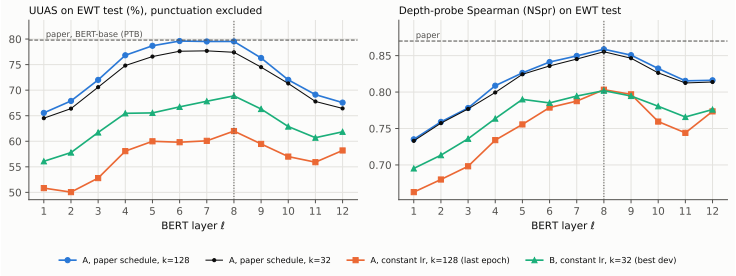
- WordSim correlation is near zero when all 353 pairs are included.
- Google analogy accuracy remains below 1% on covered questions.
- MST scores exceed random trees but remain far below a linear chain.

Why covered ρ differs

B's 203 covered pairs exclude rare words. On this same subset, **A**'s Task 1 vectors score 0.171; this is a coverage effect, not evidence of a stronger model.

Pipeline: EWT training text $\rightarrow$ word–word co-occurrence matrix $\rightarrow$ PPMI weighting $\rightarrow$ truncated SVD (100 dimensions) $\rightarrow$ cosine similarity / vector-offset analogy / sentence MST.

Task 2: same best layer, different absolute UUAS



Layer-wise test performance. The mid-layer peak appears in both runs; the absolute level depends strongly on the optimization schedule.

$\ell^* = 8$

best syntax layer in both implementations

	UUAS	NSpr
Paper (PTB)	79.8	0.87
A, lr decay	79.5	0.859
B, constant lr	68.9	0.802
A, constant lr	62.0	0.803

Interpretation

Both independently identify layer 8. With the paper-style schedule, **A** reaches 79.5% UUAS, close to the 79.8% paper reference.

Probe setup: frozen BERT layers 1–12; linear distance and depth projections; ranks $k \in \{32, 64, 128\}$; Adam; MST decoding of predicted pairwise distances; choose ℓ^* on the dev split and report UUAS / depth-probe Spearman on test.

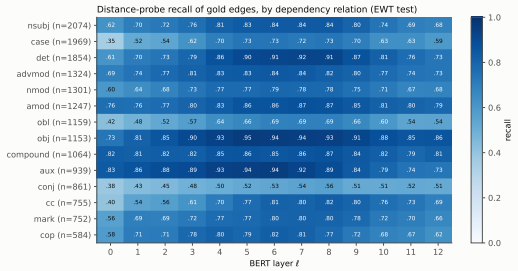
Why the UUAS differs: learning-rate schedule and relation difficulty

Evidence from the two runs

- **A**'s control with constant lr drops to 62.0% (last epoch); **B** reports 68.9% using its best-dev epoch.
- **B**'s best $k = 128$ checkpoint is mostly epoch 1. **A**'s constant-lr dev minimum is also at epoch 1, then it oscillates for 29 epochs.
- Training loss remains high (0.62 vs. 0.42 with decay): this is non-convergence, not over-fitting.
- Rank preference also flips: constant lr favours $k = 32$; decay favours $k = 128$ (77.4 / 79.2 / 79.5%), as in the paper.

Mechanism

With a constant step size, Adam keeps moving around the minimum. The paper's $\times 0.1$ decay after a dev plateau reduces the step size and lets the probe settle.

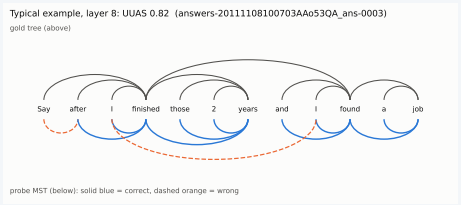


A's layer-wise recall of gold dependency edges by relation on EWT test.

What is easier / harder? Short local relations such as object (0.93), determiner (0.91), and auxiliary (0.89) are recovered well. Coordination (conj, 0.56) and oblique attachment (obl, 0.69) are harder.

Task 2: what the probe gets right – and where it fails

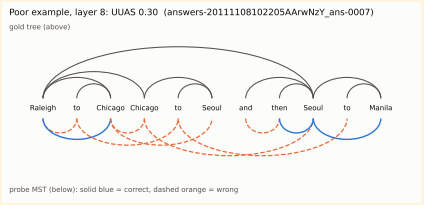
Typical test sentence: UUAS = 0.82



Reading: short web sentence; most local arcs are recovered, but discourse and coordination links are fragile.

Observed errors: misses the discourse attachment Say–finished and the conjunct link finished–found; adds Say–after and I–I.

Poor test sentence: UUAS = 0.30



Reading: repeated city names and prepositions cause lexical-identity links to compete with syntactic relations.

Observed errors: a verbless itinerary with repeated word forms; the probe creates spurious links such as to–to and Chicago–Chicago.

What these examples reveal

Layer-8 BERT representations contain useful syntactic geometry, but similar lexical forms can still dominate pairwise distances. The probe is strong in aggregate, not uniformly correct sentence by sentence.

Task 3, meaning: both find a large gain

	System	WS ρ (203 pairs)	Analogy (covered)
A	T1 PPMI–SVD	0.171	0.51%
B	T1 PPMI–SVD	0.189	0.85%
A	T3 BERT-static, L8	**0.648**	**67.7%**
B	T3 BERT-static, L8	**0.648**	**77.1%**
A	T3 + all-but-the-top	**0.716**	71.0%
B	T3 mean-centred	**0.706**	78.3%

Statistical evidence (A)

$\Delta\rho = +0.474$ with 95% CI [0.36, 0.61]. T3 solves 3,848 analogy questions missed by T1; the reverse count is 0.

Why the gain is large

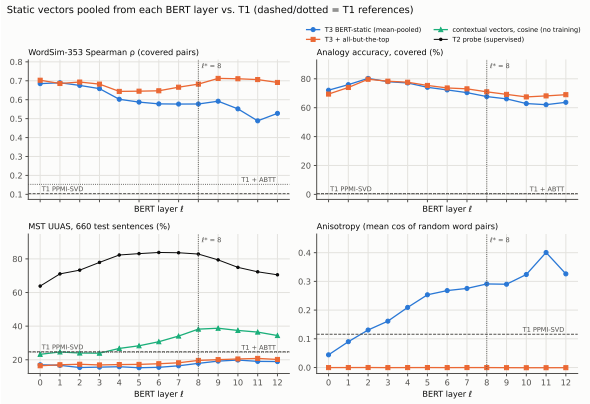
- WordSim ρ rises from about 0.18 to 0.65 on the matched 203 pairs.
- Covered analogy accuracy moves from below 1% to 68–77%.
- Removing the dominant shared direction improves T3 in both implementations.
- B's analogy candidate set is restricted to 8,865 words, raising its covered accuracy.

Important comparison caveat

This is not a data-matched pre-training comparison: BERT already learned from approximately 3.3B words; EWT supplies the contexts used for pooling.

Layer distinction: the best syntax layer is 8, but semantic static-vector quality peaks earlier; the full layer sweep is shown next.

Task 3, layer sweep: the best layer depends on the target



Word meaning

WordSim-353 peaks at layer 1 (A: 0.690; B: 0.723 on the matched pair set).

Analogical structure

Google analogy accuracy peaks at layer 2 (A: 80.3%; B: 86.2%).

Syntax / tree check

Probe UUAS peaks at layer 8; static MST is best around layer 10 but remains near chance.

Task 3, trees: both find no improvement

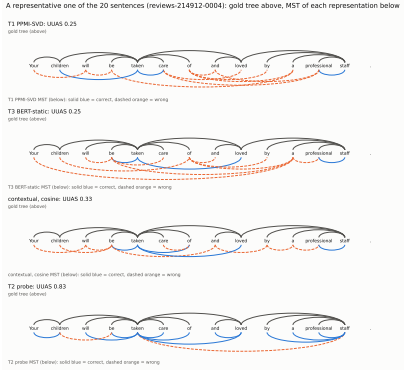
UUAS (%)	20 sentences A	20 sentences B	larger set A (660)	larger set B (1,926)
Random tree	15.2	13.1	17.1	14.3
Linear chain	43.2	35.5	43.0	40.5
T1 PPMI–SVD	20.5	16.7	24.8	17.8
T3 BERT-static	**18.4**	**9.6**	**17.9**	**14.5**
T3 + anisotropy fix	19.7	13.5	19.6	16.0
T2 probe (A)	81.2	–	82.9	–

Result across both implementations

T3 is below T1 in both samples and remains close to the random-tree baseline. Better semantic similarity does not translate into better sentence-tree recovery.

Significance (A)

20-sentence change: Δ =-2.1 pt, not significant. 660-sentence change: Δ =-6.9 pt, CI $[-8.0, -5.8]$.



Representative EWT sentence: T1, T3, contextual cosine, and the learned probe yield very different trees.

Why meaning and syntax disagree: from word types to token relations

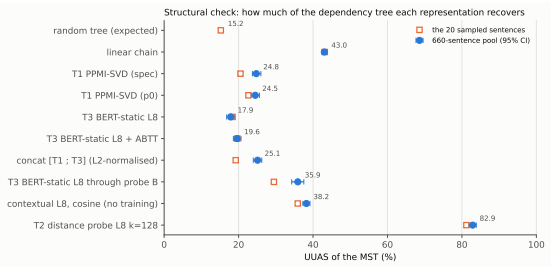
Explanation from implementation B

- Pooling averages away context-specific information.
- The structural probe learns a projection; the MST check uses raw cosine distances.
- BERT-static neighbours act like semantic substitutes, while syntax depends on relations between words in context.
- OOV words can inject arbitrary distances into B's MST sample (7.4% of tokens).

A's quantitative diagnostics

On T3, 0.74 of nearest neighbours share the same POS (T1: 0.33; random: 0.22). T1 neighbours co-occur 0.56 of the time (T3: 0.16; random: 0.07).

Interpretation: static neighbours are semantic substitutes; dependency edges connect contextual partners.



Same 660-sentence sample: each step restores a more syntax-aware signal.

Does geometric closeness predict a gold edge?

Edge AUC: T3 static **0.46**; T1 static 0.54; contextual cosine 0.70; learned probe **0.96**. The trained projection is much more syntax-aligned than raw static-vector geometry.

What survives a change of implementation?

Robust: both implementations agree

- $\ell^* = 8$: syntax is strongest in the middle layers, then declines toward layer 12.
- T3 $\gg$ T1 on WordSim-353 and analogies.
- T3 $<$ T1 on the MST check; tree recovery remains close to random.
- Static semantics peaks at layers 1–2, not at syntax layer 8.
- Centering / all-but-the-top improves T3 semantic quality.
- T3 numbers match exactly once vocabulary and evaluation settings are aligned.

Implementation-sensitive: raw values move

- Probe UUAS ranges from 62% to about 80% with learning-rate schedule / checkpoint choice.
- Best probe rank changes: 32 vs. 128.
- T1 WordSim ρ changes from 0.10 to 0.17–0.19 with pair subset and vocabulary cutoff.
- MST levels differ by up to 8 points with sentence sample and punctuation.
- "All 353" ρ depends on the OOV imputation rule.

Reproducibility lesson: report vocabulary thresholds, benchmark coverage, OOV handling, punctuation policy, probe rank, checkpoint rule, and learning-rate schedule. These choices can move the reported number more than the effect being investigated.

Conclusions

T1	T2	T3
PPMI–SVD baseline	Structural probe	BERT-static, layer 8
$\rho \approx 0.1$–0.19	**79.5%**	$\rho = 0.648$
Analogies below 1%	**UUAS with paper-style schedule**	**67.7–77.1% analogy accuracy**

1. **Task 1:** classical vectors from only 204k training tokens are weak and data-limited in both implementations.
2. **Task 2:** BERT encodes dependency structure in a low-rank subspace of its middle layers; both identify $\ell^* = 8$.
3. **Task 3:** pooling layer-8 contextual vectors greatly improves word meaning, but weakens dependency-tree recovery compared with Task 1.
4. The disagreement is not a contradiction: type-level semantic similarity and token-level syntactic relations are different targets.
5. Two independent implementations make the conclusions more robust and reveal which experimental choices are sensitive.

A good static meaning vector is not automatically a good syntactic representation.

References

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- J. Mu and P. Viswanath (2018). *All-but-the-top: simple and effective postprocessing for word representations*. ICLR.
- K. Ethayarajh (2019). *How contextual are contextualized word representations?* EMNLP.
- M. Sahlgren (2006). *The Word-Space Model*. PhD thesis, Stockholm University.
- M.-C. de Marneffe et al. (2021). *Universal Dependencies*. Computational Linguistics, 47(2).

Project materials

Combined report: `Combined/report_combined.pdf`; individual implementation reports and experiment outputs in each author's folder; code repository: `https://github.com/trishitasaha/NLP_Assignment2`.

Benchmark results and implementation comparisons in this presentation are taken from the supplied combined report. All A/B differences are interpreted with the report's vocabulary, OOV, sample-size, and punctuation caveats.